LLL-D-581D
July 9, 1969
SUPERSEDING
Int. Fed. Spec. LLL-D-00581C(GSA-FSS)
September 17, 1968, and
Fed. Spec. LLL-D-581B
November 7, 1966

FEDERAL SPECIFICATION

DOORS, WOOD, (EXTERIOR AND INTERIOR)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers exterior and interior wood doors, and specialty doors which offer sound insulation, resistance to fire, and attenuation of gamma and x-rays.

1.2 Classification.

1.2.1 Types and classes. Doors shall be of the following types and classes, as specified (see 6.2):

Type I - Hardwood veneered.
Type II - Softwood (solid and veneered).
  Class A - Ponderosa pine.
  Class B - Fir, spruce, and hemlock.
Type III - Fire resistant.
Type IV - Sound resistant.
Type V - X-ray attenuated.
Type VI - Plastic overlay, flush.
  Class C - Film faced.
  Class D - Laminated, thermosetting.
Type VII - Hardboard.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

DD-G-451 - Glass, Plate, Sheet, Figured (Float, Flat, For Glazing, Corrugated, Mirror and Other Uses).
MMM-A-260 - Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-C-650 - Crates, Wood, Open and Covered.
PPP-C-843 - Cushioning Material, Cellulosic.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. Department of Commerce Commercial (CS) and Products (PS) Standards:

CS 35 - Hardwood Plywood.
CS 73 - Old Growth Douglas, Fir, Sitka Spruce, and Western Hemlock Standard Stock Doors.
CS 120 - Ponderosa Pine Doors.
CS 171 - Hardwood Veneered Doors (Solid Core, Hollow Core, and Panel and Sash).
CS 236 - Mat-Formed Particle Board.
CS 262 - Water Repellent Preservative Non-Pressure Treatment for Millwork.

PS 1 - Softwood Plywood, Construction and Industrial.

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

American Society for Testing and Materials (ASTM) Standards:

C165 - Test for Compressive Strength of Preformed Block-Type Thermal Insulation.
C203 - Test for Breaking Load and Calculated Flexural Strength of Preformed Block-Type Thermal Insulation.
C303 - Test for Density of Preformed Block-Type Thermal Insulation.
D1037 - Evaluating the Properties of Wood-Base Fiber and Particle-Panel Materials.
D1554 - Definition of Terms Relating to Wood-Base Fiber and Particle-Panel Materials.
E90 - Recommended Practice for Laboratory Measurement of Airborne-Sound Transmission Loss of Building Partitions.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103.)

Underwriter's Laboratories Inc. (UL) Standard:

UL 10(b) - Fire Test for Door Assemblies.

(Applications for copies should be addressed to the Underwriter's Laboratires, Inc., 207 East Ohio St., Chicago, Ill. 60611; 1655 Scott Blvd., Santa Clara, Calif. 95050; 2550 Dundee Rd., Box 247, Northbrook, Ill. 60662; 1285 Walt Whitman Rd., Melville, N. Y. 11749.)

3. REQUIREMENTS

3.1 Interior and exterior doors. Doors shall be suitable for interior or exterior mounting, as specified (see 6.1 and 6.2).

3.2 Treatment. When specified (see 6.2 and 6.7), exterior doors shall be treated with water-repellent preservative in accordance with CS 262.

3.3 Glazing. When specified (see 6.2), doors shall be glazed with a glass conforming to DD-G-451. All glazed exterior doors, except combination storm and screen doors, shall have the glass bedded in putty or glazing compound and the wood beads nailed securely in place.

3.4 Finish. All types of doors shall be unfinished, prefitted, prefinished on faces and edges, mortised for locks, and cut for hinges, as specified (see 6.2).

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3.5 Type I hardwood, veneered. Hardwood veneered doors shall conform to CS 171, except that in addition to type I flush doors conforming to CS 171, flush doors with cores of mat-formed particleboard (see 3.5.3.1) or other dry processed lignocellulose materials (see 3.5.3.2) are acceptable. Doors shall be of the standard stock design, layout number, size and thickness, and grade, as specified (see 6.2).

3.5.1 Adhesives and bonds. Adhesives used for all fabrications of interior doors shall be at least equal to those used for type II plywood (water resistant bond) in CS 35 unless otherwise specified (see 6.2). Adhesives used for all glue lines between veneer plies and between the core unit and the veneer plies of exterior doors shall be at least equal to type I plywood (fully waterproof bond) in CS 35.

3.5.2 Solid wood core flush. The core shall be constructed of low density wood blocks of any length and of one species. They shall not be more than 2-1/2 inches wide. Unless otherwise specified (see 6.2), core material used for flush solid core doors shall weigh not more than 37 pounds per cubic foot, oven-dry basis.

3.5.2.1 Glued block core. The wood blocks shall be bonded together with the end joints staggered in adjacent rows unless the blocks are full length. Open spaces between the core blocks or defects in the core blocks shall not be large enough to show through the face or materially affect the appearance of the door. Stiles and rails shall be at least 1/2 inch wide after trimming and shall be bonded to the edges of the core.

3.5.2.2 Framed block glued core. Core shall be as described in 3.5.2.1 except that stiles and rails shall be at least 1-1/8 inch wide after trimming and need not be bonded to the core.

3.5.2.3 Framed block non-glued core. The core shall be as described in 3.5.2.2 with the core blocks uniformly fixed on all four edges except that core blocks shall not be bonded together but assembled so as to completely fill the frame. Minimum width of stiles and rails shall be 1-1/8 inch after trimming.

3.5.2.4 Stile and rail core. Stile and rail core shall be constructed with stile, rail, and panel units, each unit being made up entirely of blocks bonded together. In the stile and rail units, the blocks shall be laid up with the grain running parallel and in the direction of the longitudinal dimension of the stile and rail. In the panel units, when the core is to be crossbanded, the blocks shall be laid up with the grain running vertically and parallel and horizontally and parallel, when no crossbanding is used. Edge bands not less than 1/2 inch wide after trimming shall be applied on the edges of the stiles and rails, but not on the ends of the stiles.

3.5.2.5 Veneers. Veneers for crossbanding and faces combined shall be of two or more veneer plies, with a total minimum thickness of 1/2 inch before sanding. Face veneers shall be standard commercial thickness but not less than 1/36 inch before sanding. When no crossbanding is used the core shall be of stile and rail construction and the face shall be at least 3/16 inch thick before sanding.

3.5.3 Solid composition board core flush. The core shall be produced utilizing methods and controls such that the cores shall be consistently in accordance with all requirements given herein. Core may consist of one or more pieces, edges or ends glued together so long as the resultant panel meets all the requirements set forth. The core panel shall not create open spaces which would be large enough to show through the face or materially affect the strength of the door. The maximum thickness swelling shall be 3.5 percent when tested in accordance with 4.4.4.

3.5.3.1 Mat-formed wood particleboard core. The core shall be as described in 3.5.3 and shall consist of particles of wood bonded together with a synthetic resin or other suitable binder, by means of controlled production methods. In general, the methods include classifying the particles by size and drying them to a uniform moisture content. Subsquently, they are mixed with a binder, mat-formed into a panel, compressed to proper density; and then cured under heat and pressure. Core shall conform to the property requirements for type I, density C, class I particleboard as listed in table I of CS 236.

3.5.3.2 Other mat-formed particle cores. Cores made of dry processed lignocellulose materials shall be permitted provided they meet the requirements for mat-formed wood particleboard as specified in 3.5.3.1. Core material shall be as defined in ASTM D1554.

3.5.3.3 Mineral core. The core shall consist of mineral materials bonded into the inert and rigid pieces of full core thickness. Joints shall be tight and the core shall be bonded to the stiles and rails. Mineral core shall meet the following requirements.

(a) Density (oven-dry basis): 16 to 28 pounds per cubic foot (see 4.4.5).
(b) Normal tensile strength (ultimate): Minimum of 30 p.s.i. determined according to 4.4.6.
(c) Flatwise compressive strength: Minimum of 200 p.s.i., determined according to 4.4.7.

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3.5.3.3.1 Accelerated aging cycles. Material shall be capable of withstanding six complete aging cycles in accordance with 4.4.8, without major distintegration.

3.5.3.3.2 Moisture absorption by weight, based on oven dry condition shall be not more than 10 percent when tested in accordance with 4.4.9.

3.5.3.3.3 Modulus of rupture shall be at least 125 p.s.i. when tested according to 4.4.10.

3.5.4 Edge bands. Edge bands shall be smooth, straight cut, and shall be free from knots, pitch pockets, and other defects for a distance of at least 1/2 inch from the outside edge.

3.5.4.1 Style 1. The minimum width of edge bands shall be 1-1/8 inch after trimming when not glued to the core. When edge bands are glued to the core under pressure, the width of the side edge banding may be reduced to a minimum of 1/2 inch after trimming provided adequate fastening of hinges is obtainable.

3.5.4.2 Style 2. Edge bands shall be glued to the core under pressure. The side edge bands shall be at least 1-3/8 inch wide and the top and bottom edge bands shall be at least 4-3/8 inches wide. Wood lock-blocks 3-13/16 inches wide and 35-5/8 inches long shall be provided on both side edges of the door.

3.6 Type II, softwood (solid and veneered).

3.6.1 Class A, ponderosa pine. Class A doors shall conform to CS 120 except that, in addition to doors conforming to CS 120, panels for panel and sash doors and cores for flush doors may be of the composition board construction as specified in 3.5.2 and 3.5.3. Exterior use doors shall have exterior type adhesive bond. Interior use doors shall have either interior or exterior type adhesive bond, at the option of the manufacturer.

3.6.2 Class B, fir, spruce, and hemlock. These doors shall conform to CS 73. Exterior use doors shall have exterior type adhesive bond conforming to PS 1. Interior use doors shall have either interior or exterior type adhesive bond conforming to PS 1 at the option of manufacturer.

3.6.3 Designs, dimensions, and grades. Type II doors shall be of the standard stock design, layout number, size, thickness, and grade, as specified (see 6.2).

3.6.4 Flush hollow-core doors. The doors shall be manufactured in accordance with the hollow-core flush veneered door requirements of CS 171. Softwood veneers shall conform to PS 1.

3.7 Type III, fire resistant. Fire-resistant doors, shall comply with the applicable requirements for solid-core flush doors, interior use, as specified (see 3.5 and 3.6.1), and in addition, shall satisfy the fire resistance test requirements as described in UL 10(b), class B 1 hour or 1-1/2 hour, or class C 3/4 hour, as specified (see 4.2 and 6.2). When specified, (see 6.2), each contractor shall submit to the contracting agency proof that the doors he proposes to supply under this specification conform to the test requirements of UL 10 (b). The label of the UL, Inc., may be accepted as evidence that the doors conform to this requirement.

3.8 Type IV, sound resistant. Sound-resistant doors shall comply with the applicable requirements for solid-core flush doors, interior use, as specified herein (see 3.5 and 3.6.1), and in addition, shall effectively reduce sound to 35, 40, or 43 decibels, as specified (see 4.4.3 and 6.2).

3.9 Type V, x-ray attenuation. X-ray attenuated doors shall comply with the applicable requirements for solid-core flush doors, interior use, as specified herein (see 3.5 and 3.6.1), and in addition shall have one or more continuous sheets of lead to make up a total thickness as specified (see 6.2), but in no case less than 1/32 inch thick, built in to the door. The lead sheet may be in the core, between the core and cross-banding, or between the cross-banding and the face veneers, at the option of the manufacturer; but each sheet shall be continuous from edge to edge, and from top to bottom, and must be an integral part of the door. The faces of doors faced with hardwood veneers, hardboard, or plastic shall be as specified (see 6.2), and as specified, shall conform to the face grading requirements of CS 171.

3.10 Type VI, plastic overlay, flush.

3.10.1 Class A film faced. The faces of class A film faced doors shall comply with the applicable requirements of PS 1 for overlaid plywood. The doors shall be manufactured in accordance with the solid-core or hollow-core flush door requirements of CS 171, except that, in addition to doors conforming to CS 171, doors with core of composition board construction shall be acceptable, as described in 3.5.3.

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3.10.2 Class B, laminated, thermosetting. Class B laminated thermosetting plastic doors shall comply with the applicable requirements of CS 171 except that, in addition to doors conforming to CS 171, doors with cores of composition board construction shall be acceptable, as described in 3.5.3. Class B laminated thermosetting plastic shall be (1) on one or two faces, (2) on one or two faces and all edges, as specified, and in accordance with the plastic face grading requirements as specified, (see 6.2). Under condition (2) above, class B laminated thermosetting plastic shall be in accordance with the prefitting requirements of CS 171. The plastic pattern and color shall be as specified (see 6.2).

3.11 Type VII, hardboard. Doors with hardboard faces shall comply with the applicable requirements of CS 171, except that doors with cores of composition board construction shall be acceptable, as described in 3.6.1 and 3.5.3.

3.12 Warp, type I through VI, inclusive. Warp for these doors shall not exceed 1/4 inch when measured in accordance with 4.3.3.

3.13 Racking, all types. When specified (see 6.2), racking in plane of doors shall not exceed 0.1 inch, and on removal of load, permanent set shall not exceed 0.02 inch when tested in accordance with 4.4.1.1. When specified (see 6.2), racking perpendicular to plane of doors shall not exceed 3 inches at the center of the door and on removal of the load, the permanent set shall not exceed 1/8 inch when tested in accordance with 4.4.1.2.

3.14 Identification marking. Type III fire-resistant doors shall have affixed on the hinge edge the underwriters' label or, in the case of certification by the contractor, a label specifying the fire-resistance test which the contractor certifies that the door has satisfied (see 3.7 and 3.8). Type IV sound-resistant doors shall have affixed on the hinge edge a label specifying the sound-resistance test which the contractor certifies that the door has satisfied (see 3.9). Types I, II, III, VI and VII doors shall have affixed a label or permanent marking identifying or describing the door type and class, adhesive bond type, and in the case of hardwood veneered flush doors, the face grade, the face wood veneer grade, if plastic or hardwood. The terminology used to describe types III and IV doors, other than the description of special fire or sound resistant characteristics, and the terminology used to describe types I, II, VI, and VII doors shall agree with the applicable terminology used in CS 171 for hardwood veneered doors; CS 120 for softwood, ponderosa pine doors; or CS 73 for softwood, fir, spruce, and hemlock doors. Exterior doors treated with water repellent preservative shall have a label affixed or certification that the doors comply with the requirements of CS 262. In addition to the marking and certification requirements of CS 171, exterior hardwood veneered flush doors may be identified for fully waterproof bond by a plastic plug located in the rail of the door within 2 inches of the joint of the stile and rail.

3.15 Workmanship. Doors shall be free from defects affecting appearance or as specifically provided for herein and as described in the applicable sections of CS 171 in the case of hardwood veneered flush, panel, and sash types; CS 120 in the softwood, ponderosa pine type; and CS 73 in the softwood, fire, spruce, and hemlock types.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all doors of the same type, class, and size offered for delivery at the same time.

4.2.2 Dimensional and visual examination of end item. Sampling of doors for the characteristics set forth in Table I shall be in accordance with inspection level S-3.

4.2.3 Test samples.

4.2.3.1 Types III and IV. Samples of fire-resistant and sound-resistant doors, unless previously tested and acceptable for certification, shall be 2 doors for each type.

4.2.3.2 Types I, II, V, VI, and VII. The number of these doors for tests shall be selected in accordance with level S-2.

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4.2.3.3 Types I to VII. The acceptable quality level (AQL) shall be 4.0 percent defective for all tests, and for each test where applicable.

4.3 Examination.

4.3.1 Defects. Doors selected in accordance with 4.2.2 shall be examined in accordance with table I. The AQL's for defects shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects in accordance with MIL-STD-105.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Type of door not as specified	X	
Class, grade, and thickness of doors not as specified	X	
Design, layout, or size of doors not as specified	X	
Doors not preservative treated as specified	X	
Doors not glazed as specified	X	
Species of wood not as specified	X	
Fire- or sound resistant doors not labeled as required	X	
Thickness of lead for x-ray doors not as specified	X	
Broken or cracked glass		X
Damage or defects affecting function or serviceability	X	
Damage or defects not affecting function or serviceability		X

4.3.2 Thickness of doors. Doors selected in accordance with 4.2.2 shall be measured using a rule graduated in 1/32 inch increments.

4.3.3 Measurement of warp. Doors selected in accordance with 4.2.2 shall be measured from the door face to the center of a straight edge positioned to determine the maximum amount of bow, cup, or twist (see 3.12).

4.3.4 Examination of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

4.4 Test methods.

4.4.1 Racking, all types. One door shall be tested from the samples selected in 4.2.3 (see 3.13).

4.4.1.1 In plane of doors. Support the door rigidly as in service. Apply a load of 200 pounds to the outer stile or edge of the door.

4.4.1.2 Perpendicular to the plane of the door. Support the door horizontally on two diagonally opposite corners. Apply a load of 100 pounds to each of the other corners. Ambient temperature and humidity shall be 72° ± 3° F. and 50 ± 3 percent relative humdity.

4.4.2 Fire resistance, type III. Test shall be conducted as described in UL 10(b) (see 3.7).

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4.4.3 Sound resistance expressed as the average transmission loss in decibels over a range from 128 to 4,096 cycles, shall be determined in accordance with ASTM E 90, modified by mounting the door in a frame equipped with gasketed stops (see 3.8).

4.4.4 Linear expansion. Determine linear expansion in accordance with CS 236 (see 3.5.3).

4.4.5 Density. Determine density in accordance with ASTM C 303 (see 3.5.3.3(a)).

4.4.6 Normal tensile strength (ultimate). Determine normal tensile strength in accordance with sections 27 to 31 of ASTM D 1037 (see 3.5.3.3(b)).

4.4.7 Flatwise compressive strength. Determine flatwise compressive strength in accordance with sections 2 to 6 of ASTM C 165 (see 3.5.3.3(c)).

4.4.8 Accelerated aging. Determine accelerated aging in accordance with sections 82 to 84 of ASTM D 1037 (see 3.5.3.3.1):

4.4.9 Moisture absorption. Test for moisture absorption in accordance with sections 69 to 75 of ASTM D 1037 (see 3.5.3.3.2).

4.4.10 Modulus of rupture. Test in accordance with ASTM C 203 (see 3.5.3.3.3).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each door shall be packaged in a close fitting box conforming to PPP-B-636, W5c, or W6c, style optional.

5.1.2 Level B. Same as A except fiberboard shall be type CF, class domestic.

5.1.3 Level C. The doors shall be packaged to afford protection against deterioration and damage during handling and shipment from contractor to the initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The doors of like description, packaged as specified in 5.1, shall be packed in a close-fitting box, conforming to PPP-B-601, overseas type, style optional. The box shall be closed and strapped in accordance with the appendix to the box specification. The strapping shall be zinc coated.

5.2.2 Level B. Doors of like description, packaged as specified in 5.1, shall be packed together as specified in 5.2.1 for level A, except domestic type box shall be used and strapping need not be zinc coated; or packed together in a close fitting box conforming to PPP-B-640, class 2, style optional. The gross weight of size shall not exceed the limitations of the applicable box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. Doors of like description packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

6. NOTES

6.1 Intended use. All types of doors are intended for interior or exterior use. Fire-resistant, sound-resistant, and x-ray attentuated doors are usually intended for interior use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of doors (see 1.2.1).
(c) Whether for interior or exterior mounting (see 3.1).
(d) Whether treatment is required (see 3.2 and 6.7).
(e) Whether glazing is required (see 3.3).
(f) Finish (see 3.4).
(g) Design, layout number, size, thickness, and grade for type I (see 3.5).
(h) If other than type II plywood (water resistant bond) is required (see 3.5.1).
(i) Whether solid core material shall exceed 37 pounds per cubic foot (see 3.5.2).
(j) For all types, size and thickness, grade, whether of panel, sash, or flush construction, and the facing material or wood species (see 3.5, 3.6.3 and 3.7 through 3.11).

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(k) Design and layout number for type II doors (see 3.6.3).
(l) Class of fire resistance for type III doors (see 3.7).
(m) Level of sound reduction for type IV doors (see 3.8).
(n) Total thickness of lead required for type V doors (see 3.9).
(o) Overlay pattern, and color for laminated thermosetting plastic doors, type VI, class B doors (see 3.10.2).
(p) Whether racking tests are required (see 3.13)
(q) Whether label is required for type III doors (see 3.4 and 6.8).
(r) Size of lot, if different from 4.2.1.
(s) Selection of applicable levels of packaging and packing required (see 5.1, 5.2, and 6.4).

6.3 Detailed information on standard stock designs and layouts, standard sizes and tolerances, materials, grading, and construction are shown in CS 73, CS 120, and CS 171. For types I, III, IV, V VI and VII, see CS 171. For type II see CS 73 and CS 120.

6.4 For civil agency procurements. When level B packaging is required for civil agencies procurements, the level A requirements in 5.1 shall apply.

6.5 Hardware. Locks and door trim for doors are covered by FF-H-106, Hardware, Builders'; Locks and Door-Trim.

6.6 Classification of types. Type II doors in LLL-D-581 is type III in this specification.

6.7 Treatment. Water-repellent preservative treatment, when required, is generally given to surfaces to be painted. Hardwood doors normally are given stained or natural finishes, and treatment of these wood species may impair their appearance. It is recommended that only when hardwood doors are to be subjected to extreme environmental conditions that application of preservatives be considered.

MILITARY CUSTODIANS:

Army - MO (ERDL)
Navy - YD
Air Force - 84

Preparing activity:

GSA-FSS

Review activities:

Army - MO
Navy - YD
Air Force - 84

User interest:

MC

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GENERAL SERVICES ADMINISTRATION

OFFICIAL BUSINESS

POSTAGE & FEES PAID
GENERAL SERVICES ADMINISTRATION

General Services Administration
Federal Supply Service (FMSH)
Washington, D. C. 20406

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GSA FORM FEB 69 2200

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

LLL-D-581D Doors, Wood, (Exterior And Interior)

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

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